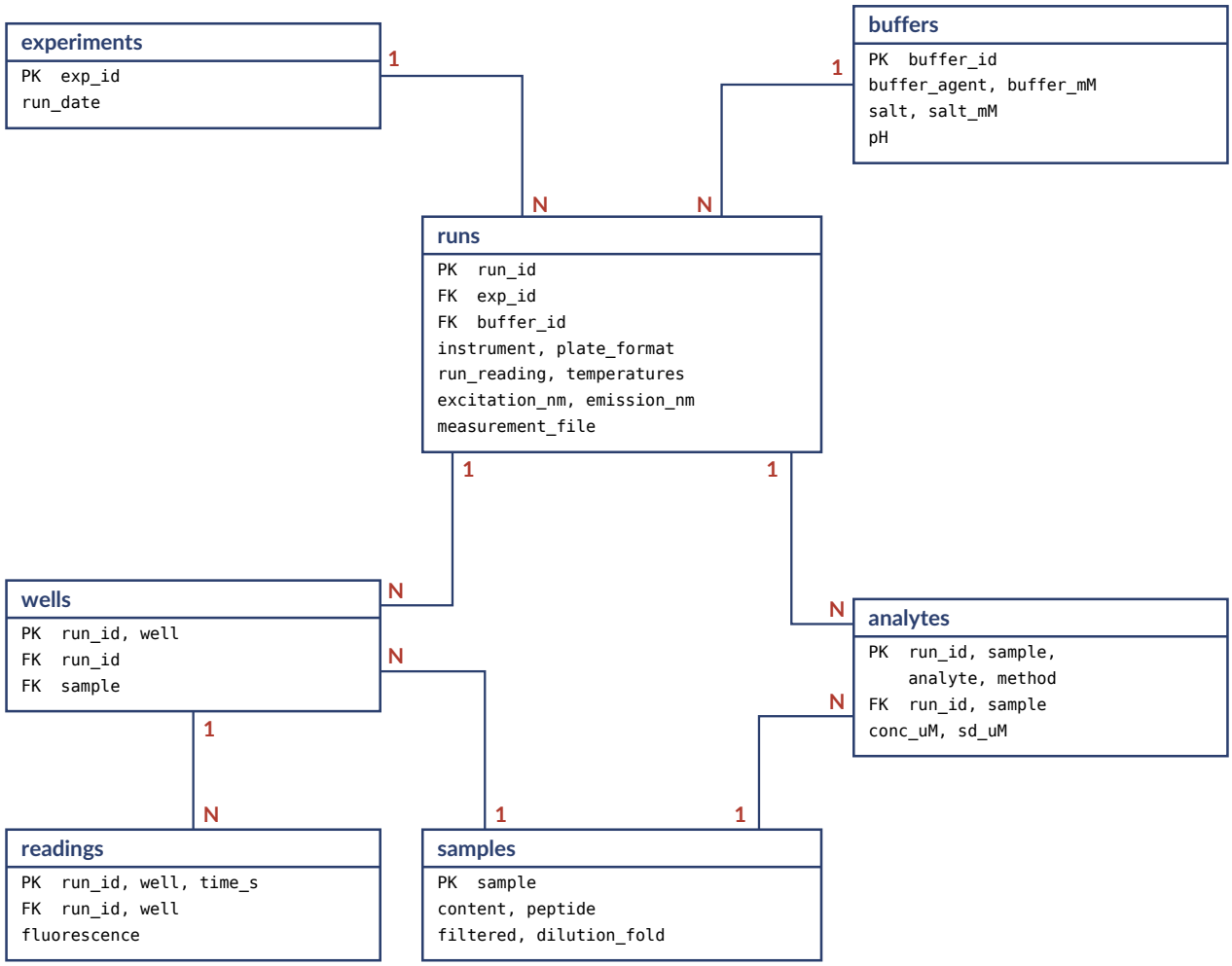
Plate-reader Kinetic Assay Database

Logical model, relationships and business rules, data dictionary, source-to-target mapping and load order

SQL (SQLite) reconstruction of the data model I used in the lab to keep 500+ plate-reader datasets collected over ten years in one place. The SQL schema and the loading and cleaning scripts for three plate-reader formats are on GitHub (github.com/heliosight/plate-reader-sql).

LOGICAL MODEL



PK = primary key, FK = foreign key. Lines show one-to-many links, with 1 on the parent side and N on the child side.

RELATIONSHIPS AND BUSINESS RULES

Parent	Child	Link	Meaning and business rule
experiments	runs	1 : N	An experiment groups one or more plate runs (runs.exp_id).
buffers	runs	1 : N	Each run is performed in one buffer composition (runs.buffer_id).
runs	wells	1 : N	A run has one record per occupied well (key: run_id + well).
samples	wells	1 : N	Each well holds exactly one defined sample; a sample can be placed in many wells.
runs, samples	analytes	1 : N	Measured concentrations are recorded per run and sample, by analyte and method.
wells	readings	1 : N	One fluorescence value per well and time point (key: run_id + well + time_s).
Business rules	all		Value ranges and allowed lists are enforced by the schema. Salt and salt concentration are set together or not at all. Each source file belongs to exactly one run.

DATA DICTIONARY

Table	Field	Type	Key	Definition and rules
experiments	exp_id	TEXT	PK	Experiment identifier.
	run_date	TEXT		Date of the experiment. Required.
buffers	buffer_id	TEXT	PK	Buffer composition identifier.
	buffer_agent	TEXT		Buffering agent. Required.
	buffer_mM	REAL		Buffer concentration, mM. Required.
	salt	TEXT		Added salt, optional.
	salt_mM	REAL		Salt concentration, mM. Set if and only if salt is set.
	pH	REAL		Required; between 0 and 14.
samples	sample	TEXT	PK	Sample label.
	content	TEXT		One of: protein+ThT, ThT, buffer, water, empty. Required.
	peptide	TEXT		Peptide used, where applicable.
	filtered	INTEGER		0 or 1.
	dilution_fold	INTEGER		Between 1 and 64.
runs	run_id	TEXT	PK	Plate run identifier.
	exp_id	TEXT	FK	References experiments. Required.
	run_reading	TEXT		Read direction: top or bottom. Required.
	instrument	TEXT		Plate reader used; selects the parser at load time. Required.
	plate_format	INTEGER		One of 24, 96, 384, 1536. Required.
	plate_temp_start_c	REAL		Plate temperature at start, 0 to 99 C. Required.
	temp_run_c	REAL		Run temperature setpoint, 0 to 99 C. Required.
	excitation_nm	INTEGER		Excitation wavelength, 220 to 900 nm. Required.
	emission_nm	INTEGER		Emission wavelength, 220 to 900 nm. Required.
	measurement_file	TEXT	UNIQUE	Source export file; unique, so one file cannot serve two runs.
	buffer_id	TEXT	FK	References buffers. Required.
wells	run_id	TEXT	PK, FK	References runs.
	well	TEXT	PK	Well position (A1 to H12 on a 96-well plate).
	sample	TEXT	FK	References samples. Required.
analytes	run_id	TEXT	PK, FK	References runs.
	sample	TEXT	PK, FK	References samples.
	analyte	TEXT	PK	One of: Abeta42, ThT.
	method	TEXT	PK	Quantification method. Required.
	conc_uM	REAL		Concentration, uM; 0 or greater. Required.
	sd_uM	REAL		Standard deviation, uM; optional; 0 or greater.
readings	run_id, well	TEXT	PK, FK	Composite reference to wells.
	time_s	INTEGER	PK	Seconds since start of run; 0 or greater.
	fluorescence	REAL		Fluorescence intensity. NULL allowed for blank or saturated reads.

SOURCE-TO-TARGET MAPPING: READINGS TABLE

The three instruments export the same information in three different layouts. Each parser maps its layout onto one long format: one row per run, well and time point.

Target field	Tecan export	Synergy export	BMG export	Transformation rule
readings.run_id	runs.run_id, looked up via runs.measurement_file	same	same	Run identity comes from the curated run metadata, not from the file contents.
readings.time_s	Rows labelled like "120s"	Rows with clock time 0:01:00	Column headers like "1 h 5 min"	Convert every format to integer seconds. BMG: drop any cycle whose time label jumps backward (a mid-run "0 h" column that duplicates the previous cycle).
readings.well	Columns A1 to H12 (96 wells side by side)	Columns A1 to H12	Row labels A01 to H12 (transposed grid)	Normalize to letter + number without zero padding (A01 becomes A1).
readings.fluorescence	Cell at (time row, well column)	Cell at (time row, well column)	Cell at (well row, time column)	Reshape the wide grid to one row per (run, well, time). Non-numeric values (overflow flags, blanks) become NULL.
(not loaded)	Per-timepoint temperature column	none	none	Not stored; the schema keeps only start and run temperatures at run level.

LOAD ORDER AND SOURCE FILES FOR ALL TABLES

Step	Target table	Source file	Rule
1	experiments, buffers, samples, runs	Curated metadata files, one per table (meta/*.csv)	Loaded first so that foreign keys resolve.
2	wells	Plate layout files, one per run (wells/*.csv)	Loaded after runs and samples.
3	analytes	Curated analyte file (meta/analytes.csv)	Loaded after runs and samples.
4	readings	Instrument exports named in runs.measurement_file (readings/)	Parser chosen by runs.instrument (Tecan, Synergy or BMG). Missing files are reported and skipped; constraint violations stop the load.

Data quality is enforced at load time by the schema: value ranges, allowed value lists, paired fields, uniqueness of source files and referential integrity between all tables.